

25 — Container vessel: L3 variation 1 500 kW from the lookup

Proves · A Container vessel picking up its 1 500 kW Level 3 variation from the lookup table.

Mechanics this scenario turns on

- When the Level 3 variation field is empty the backend looks it up from the vessel type (Bulk 250 · Container 1 500 · LNG 1 000 · otherwise the 500 default), all from `appsettings.json`.
- Level 3 (DRC) damps the hotel/mission swing: $\text{variation} \times 0.8$, minus whatever the battery already shaved, so the same kilowatt is never counted twice.

Panels described below · How the variation is chosen · What 1 500 kW does downstream

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 14.

`CalculatorSettings.VesselVariations` holds three entries. Only Bulk Carrier (250) had a scenario; this adds Container, and 33 adds LNG.

How the variation is chosen

`Level3DrcService.GetVesselVariation` runs three steps:

1. Explicit `hotelLoadVariationKw` on the input wins outright. **This scenario omits the field.**
2. Exact match on the vessel type name.
3. Substring match — "Container 5,000 TEU" contains "Container" → **1 500 kW**.
4. Otherwise the 500 kW default (scenario 26).

What 1 500 kW does downstream

variation	1 500.0 kW
reduced ($\times 0.80$)	1 200.0 kW
L3 savings	142.79 t/yr

Compare with scenario 26 — the same plant, the same everything, only the variation differs:

	variation	L3 savings
26 (default)	500	45.25
25 (Container)	1 500	142.79

Three times the swing, **3.16×** the saving. Slightly super-linear, because DRC monetizes the *curvature* of the SFOC curve and a wider swing reaches further into the steep low-load region.

That relationship is the real assertion here: a lookup that silently returned the default would drop this scenario from 142.79 to 45.25 — a 97 t/yr difference no one could miss.

Takeaway: the vessel-type lookup is a live input to the numbers, not a label. It deserves a scenario per entry, and now has one.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)